

Lucas Lange

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Education

- 10/2023 - Present **Technical University of Munich (Munich, Germany)**
M.Sc. Computer Science
Focus: Machine Learning, Quantum Computing, Databases
Erasmus Semester at University of Agder (UiA), Norway
- 10/2018 - 09/2023 **Technical University of Munich (Munich, Germany)**
B.Sc. Computer Science with a minor in Psychology

Experience

- 05/2026 - Present **Agentic AI Engineering (Master's Thesis) at BMW Group**
- Architected & benchmarked 9 agent architectures from scratch - single-agent (Google ADK), hierarchical multi-agent (3 granularities), state-graph (LangGraph), deterministic-leader, plan-then-execute, and self-verification - connected via A2A protocol and MCP, enabling the first empirical architecture comparison for tool-calling enterprise agents
 - Designed a 7-track, staged-elimination experimental pipeline (model selection → prompt tuning → architecture comparison → deployment validation) with 800+ automated evaluation runs across 29 models, ranked via a weighted composite score (quality, judge, efficiency, cost) routed through LiteLLM
 - Built a dual LLM-as-judge scoring system (GPT-4o + Claude Sonnet 4.5) combined with deterministic tool-call evaluators (precision, recall, order, reliability), running in an async Python pipeline with scenario-driven test fixtures for fully reproducible benchmarking
- 10/2025 - 01/2026 **Technical Team Lead – DevOps & Infrastructure at Adtran**
- Led a 3-person engineering team within a 10-developer project, setting technical priorities, reviewing PRs, and coordinating architecture and stakeholders
 - Built the DevOps infrastructure from scratch, including CI/CD pipelines and Docker deployment of three containers to Adtran's on-premises servers
- 09/2024 - 12/2024 **Software Engineer Intern at BMW Group**
- Developed a Python-based anomaly detection system with REST APIs, collaborating with UI designers on FIGMA prototypes for the frontend interface, achieving a 10% error rate reduction in production configurations
 - Engineered key components of an intelligent search algorithm using Java Quarkus, Python, and Kafka Streams, while implementing CI/CD pipelines in a joint BMW-Microsoft team, contributing to a solution achieving €9M annual cost savings
- 03/2024 - 08/2024 **Machine Learning Working Student at OrthoCoPilot**
- Designed and implemented a RAG pipeline (LangChain, Llama 2) to extract and structure medical information from clinical documentation, collaborating with medical professionals to ensure clinical accuracy
- 09/2023 - 12/2023 **Machine Learning Working Student at UniCredit**
- Built data wrangling and preprocessing pipelines for a sentiment analysis dataset, aligning deliverables directly with stakeholders (Python, data pipeline, NLP)
- 11/2022 - 06/2023 **Data Analytics and Machine Learning Working Student at Nebumind**
- Automated an image-based anomaly detection pipeline and built the accompanying dashboard UI (Python, computer vision, dashboard)

Skills

- IT Agentic AI, LLM, RAG, Machine Learning, DL, LangChain, LangGraph, LiteLLM, OpenAI API
Python, PyTorch, Java (Quarkus), SQL, Kafka, Azure
Linux, Git, GitHub, Docker, Atlassian Suite, CI/CD
- Languages English (Native), German (Native), French (Bilingual)
- Interests Rowing, Hiking